Saddle-Dominated Capacity of the LSE Dense Associative Memory

Abstract

The log-sum-exp (LSE) Dense Associative Memory stores an exponentially large set of patterns on the N-sphere with a kernel sharpness parameter we call β_{net}. Ramsauer et al. [9] obtained a limiting storage load $\alpha_c = 1/2$ at zero temperature by modelling the inter-pattern alignments as Gaussian; their bound is independent of β_{net}. We revisit the calculation under the exact spherical alignment density. The dominant contribution to the noise floor moves from the upper edge of the alignment distribution to an interior saddle of the partition-sum integrand, with saddle value $g_{\max}(\beta_{\mathrm{net}})$. This single quantity controls every corrected boundary in the (α, T) plane. At $T = 0$ the corrected capacity is $\beta_{\mathrm{net}} - g_{\max}(\beta_{\mathrm{net}})$, the load at which the cusp of the leading-order free energy reaches the unit sphere and the retrieval basin disappears. It is finite, depends on β_{net}, sits below Ramsauer's bound at moderate β_{net}, and exceeds it at large β_{net}. Saddle-bulk Monte Carlo at $N = 100$ places the empirical basin boundary on the corrected curve, consistent with the analytical prediction.

1 Introduction

The Hopfield network [3] stores patterns as minima of a pairwise energy and retrieves the closest stored pattern from a noisy cue. Classical pairwise storage saturates at a load of about 0.138 patterns per neuron [1]. Dense Associative Memory replaces the pairwise interaction by a strongly nonlinear energy and raises the capacity to an exponential $M = \mathrm{e}^{\alpha N}$ at load α [4, 5, 2]. In the continuous version of the model, with N the dimension of the N-sphere ambient space, both the stored patterns $\boldsymbol{\xi}^{\mu} \in \mathbb{R}^N$ and the chain $\boldsymbol{x}$ live on the sphere $|\boldsymbol{x}|^2 = N$. The log-sum-exp (LSE) variant of Ramsauer et al. [9] writes

$$H(\boldsymbol{x}) = -\beta_{\mathrm{net}}^{-1} \ln \sum_{\mu=1}^{M} \exp(\beta_{\mathrm{net}}\, \boldsymbol{x} \cdot \boldsymbol{\xi}^{\mu}), \tag{1}$$

with β_{net} a kernel sharpness parameter that scales the dot product inside the log-sum-exp. β_{net} controls how strongly the energy distinguishes patterns and is independent of the simulation temperature T.

For continuous LSE, Ramsauer et al. [9] derived a limiting load $\alpha_c = 1/2$ at $T = 0$. Label $\mu = 1$ the retrieving target and $\mu \geq 2$ the spurious patterns. The argument treats the inter-pattern alignment $\varphi_{1\mu} = \boldsymbol{\xi}^1 \cdot \boldsymbol{\xi}^{\mu}/N$ (hereafter, ϕ) as a Gaussian with mean 0 and variance $1/N$, and asks at what load the typical maximum of M such alignments first reaches 1. The Gaussian extreme value satisfies $\phi_{\max}^{(G)} = \sqrt{2\alpha}$, so the condition $\phi_{\max}^{(G)} = 1$ gives $\alpha = 1/2$. The bound does not involve β_{net}.

The Gaussian is the leading term of the exact spherical alignment density near $\phi = 0$, but the densities diverge near $\phi = 1$. Whether the corrected density alters the capacity was raised by Petrova et al. [7], who derived the finite-temperature retrieval boundary $T_c(\alpha)$ for LSE and LSR under the Gaussian approximation and flagged its noise-floor derivation as the candidate failure point. Petrova [6] subsequently measured this boundary by direct Monte Carlo at moderate N for both kernels, without pursuing the low-T limit near the limiting load. The compactly supported LSR variant was analysed by Petrova et al. [8] from a different angle: by direct Monte

Carlo at $N \approx 40$–50 they showed that the apparent retrieval boundary in the simulations is a kinetic crossover. A metastable strip sits below the thermodynamic boundary, separated from the competing-pattern basins by an entropic barrier whose Kramers escape time grows exponentially in N. They note that resolving the strip at larger N would require a smarter Monte Carlo scheme than the direct method they employed, and do not provide one.

This paper revisits Ramsauer's $\alpha_c = 1/2$ for LSE under the exact spherical density, treating β_{net} as a free parameter. The corrected capacity is finite and depends on β_{net}; it sits below Ramsauer's bound at moderate β_{net}, crosses it at $\beta_{\text{net}} \approx 0.7$, and grows logarithmically with β_{net} beyond. Two boundary curves arise in the (α, T) plane: a single-pattern spinodal $\alpha_c^{\text{sp}}(\beta_{\text{net}}, T)$ at which the cusp of the leading-order free energy reaches the basin minimum and the basin disappears, and a thermodynamic crossover $\alpha_c^{\text{sd}}(\beta_{\text{net}}, T)$ at which the noise floor and the basin minimum exchange order in free energy. Both curves are set by the integrand saddle value $g_{\max}(\beta_{\text{net}})$, and they coincide at $T = 0$ at the endpoint $\beta_{\text{net}} - g_{\max}(\beta_{\text{net}})$. At $T > 0$ they separate; the spinodal is the relevant boundary for cold-initialised Monte Carlo, the saddle line is the thermodynamic crossover.

Two technical points motivate the rest of the paper. First, the naive correction (replacing the Gaussian by the exact density and applying the same right-edge logic) gives the wrong capacity: the LSE partition sum is governed by an interior saddle, not by the edge of the alignment distribution. Second, the local-refresh Monte Carlo schemes used to accelerate large-N studies fail near the capacity boundary by introducing unquenched disorder; the simulations in this paper rely on a saddle-bulk scheme in which the spurious sum is replaced by its analytic saddle value (the cusp $\phi_c(\alpha, \beta_{\text{net}})$ lies above the typical sample extreme $\phi_{\max}(\alpha) = \sqrt{1 - \mathrm{e}^{-2\alpha}}$ at every load below capacity, so no spurious pattern is retained explicitly).

Section 2 develops the theory: the Gaussian baseline, the naive exact-density variant, the interior saddle, the cusp picture, the spinodal, and the two-protocol picture. Section 3 describes the three Monte Carlo schemes (direct, truncated, saddle-bulk) used here, with a paragraph on why local refresh fails. Section 4 shows the empirical $N = 100$ heatmap together with the four boundary curves. Section 5 closes with the Ramsauer claim and the saddle-vs-spinodal distinction.

2 Theory

Setup. Both the stored patterns $\boldsymbol{\xi}^\mu$ and the chain $\boldsymbol{x}$ lie on the sphere $S^{N-1}(\sqrt{N})$. The inter-pattern alignment $\varphi_{1\mu} = \boldsymbol{\xi}^1 \cdot \boldsymbol{\xi}^\mu / N$ between a fixed retrieving pattern and a random spurious one is the projection of one uniform direction onto another; its density on $[-1, 1]$ is the spherical density

$$h(\phi) = C_N \, (1 - \phi^2)^{(N-3)/2}, \qquad C_N = \frac{\Gamma(N/2)}{\Gamma((N-1)/2)\sqrt{\pi}} \,. \tag{2}$$

At large N, h concentrates near $\phi = 0$ with variance $1/N$, but its right tail near $\phi = 1$ decays as $(1 - \phi^2)^{N/2}$, lighter than any Gaussian of matched variance. The chain-pattern alignment $\phi_\mu(\boldsymbol{x}) = \boldsymbol{\xi}^\mu \cdot \boldsymbol{x}/N$ measures the chain's overlap with pattern μ; ϕ_1 measures retrieval.

For a chain in the basin of $\boldsymbol{\xi}^1$ the LSE log-sum is dominated by the $\mu = 1$ term, $H(\boldsymbol{x})/N \approx -\beta_{\text{net}}^{-1} \ln \mathrm{e}^{\beta_{\text{net}} N \phi_1} = -\phi_1$; the β_{net} factor inside the exponential and the β_{net}^{-1} in front of the logarithm cancel for any single-term dominant contribution. Adding the perpendicular-subspace entropy $s(\phi_1) = \frac{1}{2} \ln(1 - \phi_1^2)$ and minimising $f = u - Ts$ with $u = 1 - \phi_1$ gives the retrieval free energy per spin

$$f_{\text{ret}}(T) = (1 - \phi_{\text{eq}}) - \tfrac{T}{2} \ln(1 - \phi_{\text{eq}}^2), \qquad \phi_{\text{eq}}(T) = \tfrac{1}{2}\big(-T + \sqrt{T^2 + 4}\big), \tag{3}$$

β_{net}-independent by the same cancellation.

Away from the basin the noise floor comes from the $M - 1$ spurious sum

$$S(\boldsymbol{x}) \; = \; \sum_{\mu \neq 1} \mathrm{e}^{\beta_{\text{net}} N \phi_\mu(\boldsymbol{x})}, \qquad \frac{1}{N} \ln S \; \approx \; \alpha + \frac{1}{N} \ln \int_{-1}^{1} h(\phi) \, \mathrm{e}^{N \beta_{\text{net}} \phi} \, \mathrm{d}\phi, \tag{4}$$

where we used $M = \mathrm{e}^{\alpha N}$ and replaced the sum over patterns by an integral with density h, valid in the bulk of $\boldsymbol{x}$ because the ϕ_μ are independent draws from h. The integrand exponent is $g(\phi;\beta_{\text{net}}) = \frac{1}{2}\ln(1-\phi^2) + \beta_{\text{net}}\,\phi$.

Gaussian baseline. Replacing h by the moment-matched Gaussian $\sqrt{N/2\pi}\,\mathrm{e}^{-N\phi^2/2}$ makes the exponent $-\frac{1}{2}\phi^2 + \beta_{\text{net}}\phi$ monotone in ϕ on $[-1,1]$ for $\beta_{\text{net}} \le 1$, so the integral (4) is set by the typical extreme of M Gaussian samples, $\phi_{\max}^{(G)} = \sqrt{2\alpha}$. Substituting gives $N^{-1}\ln S \to \beta_{\text{net}}\sqrt{2\alpha}$ and a noise-floor energy per spin $u_{\text{noise}}^{(G)} = 1 - \sqrt{2\alpha}$. The boundary $u_{\text{noise}}^{(G)} = f_{\text{ret}}(T)$ reads

$$\alpha_c^{\mathrm{G}}(T) = \tfrac{1}{2}\big[1 - f_{\text{ret}}(T)\big]^2, \qquad \alpha_c^{\mathrm{G}}(0) = \tfrac{1}{2}. \tag{5}$$

The β_{net}^{-1} prefactor of H and the β_{net} inside the exponent cancel each other at the right edge, so (5) is β_{net}-independent. This is Ramsauer's bound.

Naive correction: exact density at the edge. Keep the exact density h but apply the same edge logic. The typical maximum of M samples from h satisfies $M\int_{\phi_{\max}}^{1} h(\phi)\,\mathrm{d}\phi = 1$, which after computing the right tail of (2) gives $\phi_{\max}(\alpha) = \sqrt{1-\mathrm{e}^{-2\alpha}}$. The exponent at the edge is $g(\phi_{\max}) = -\alpha + \beta_{\text{net}}\,\phi_{\max}$, so $N^{-1}\ln S \to \beta_{\text{net}}\,\phi_{\max}(\alpha)$ and $u_{\text{noise}}^{(\text{ex})} = 1 - \phi_{\max}(\alpha)$. Solving $u_{\text{noise}}^{(\text{ex})} = f_{\text{ret}}(T)$ yields

$$\alpha_c^{\text{bd}}(T) \;=\; -\tfrac{1}{2}\ln\!\big[1 - (1 - f_{\text{ret}}(T))^2\big], \qquad \alpha_c^{\text{bd}}(0) = +\infty. \tag{6}$$

The boundary diverges at $T = 0$: the exact density has no finite right edge, so the edge logic returns no bound. This is the right answer for the compactly supported LSR kernel, where the kernel cuts the integral off at a finite value below 1 [8]. For LSE the edge logic is wrong whenever the integrand has an interior maximum inside $[0,1)$.

Interior saddle: the correct boundary. The exponent $g(\phi;\beta_{\text{net}})$ has derivative $g'(\phi) = \beta_{\text{net}} - \phi/(1-\phi^2)$, which vanishes at the positive root

$$\phi^*(\beta_{\text{net}}) = \frac{-1 + \sqrt{1 + 4\beta_{\text{net}}^2}}{2\beta_{\text{net}}}. \tag{7}$$

The saddle is inside $(0,1)$ for every $\beta_{\text{net}} > 0$. By Laplace's method,

$$\int_{-1}^{1} h(\phi)\,\mathrm{e}^{N\beta_{\text{net}}\phi}\,\mathrm{d}\phi \;\sim\; \sqrt{\frac{2\pi}{N\,|g''(\phi^*)|}}\;\mathrm{e}^{N\,g_{\max}(\beta_{\text{net}})}, \qquad g_{\max}(\beta_{\text{net}}) \;\equiv\; g(\phi^*;\beta_{\text{net}}) \;=\; \tfrac{1}{2}\ln(1-\phi^{*\,2}) + \beta_{\text{net}}\,\phi^*, \tag{8}$$

so $N^{-1}\ln S \;\to\; \alpha + g_{\max}(\beta_{\text{net}})$ and the noise-floor energy per spin reads $u_{\text{noise}}^{(\text{sd})} = 1 - [\alpha + g_{\max}(\beta_{\text{net}})]/\beta_{\text{net}}$. Equating this to $f_{\text{ret}}(T)$ gives the saddle-dominated boundary

$$\boxed{\alpha_c^{\text{sd}}(\beta_{\text{net}}, T) \;=\; \beta_{\text{net}}\,[1 - f_{\text{ret}}(T)] - g_{\max}(\beta_{\text{net}}).} \tag{9}$$

The dependence on β_{net} is real. At the right edge, the same β_{net} factor appears in retrieval and noise floor and cancels; that is why Ramsauer's bound is β_{net}-independent. At the interior saddle the location $\phi^*(\beta_{\text{net}})$ itself shifts with β_{net}, and the cancellation breaks. Figure 1 shows $\alpha_c^{\text{sd}}(\beta_{\text{net}}, 0) = \beta_{\text{net}} - g_{\max}(\beta_{\text{net}})$ together with Ramsauer's $\alpha_c^{\mathrm{G}} = 1/2$. $\alpha_c^{\text{sd}}(\beta_{\text{net}}, 0)$ crosses Ramsauer's flat line at $\beta_{\text{net}} \approx 0.7$: below this value the corrected capacity is more restrictive than Ramsauer's, above it more permissive. The endpoints are explicit: as $\beta_{\text{net}} \to 0$, $\phi^* \to \beta_{\text{net}}$ and $g_{\max} \to \beta_{\text{net}}^2/2$, giving the linear $\alpha_c^{\text{sd}}(\beta_{\text{net}}, 0) \to \beta_{\text{net}}$; as $\beta_{\text{net}} \to \infty$, $\phi^* \to 1 - 1/(2\beta_{\text{net}})$ and $g_{\max} \to \beta_{\text{net}} - \frac{1}{2}(\ln\beta_{\text{net}} + 1)$, giving the logarithmic growth $\alpha_c^{\text{sd}}(\beta_{\text{net}}, 0) \to \frac{1}{2}(1 + \ln\beta_{\text{net}})$. The Monte Carlo simulations of Sections 3 and 4 fix $\beta_{\text{net}} = 1$ for the numerics, where $\alpha_c^{\text{sd}}(1, 0) \approx 0.6226$.

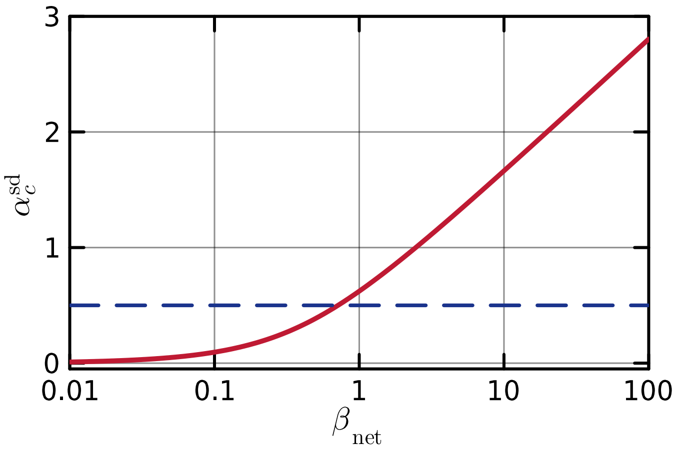

Figure 1: Zero-temperature capacity $\alpha_c^{\text{sd}}(\beta_{\text{net}}, 0)$ (9) of the LSE DAM as a function of the kernel sharpness β_{net} (red), together with Ramsauer's β_{net}-independent bound $\alpha_c^{\text{G}} = 1/2$ (5) (blue dashed).

Cusp picture. The saddle integral (8) is independent of $\boldsymbol{x}$: by isotropy of the random patterns, after averaging over the directions orthogonal to $\boldsymbol{\xi}^1$ only $\varphi_{1\mu}$ (not $\boldsymbol{x}$ itself) enters the spurious sum. The chain therefore competes inside the LSE logarithm between the retrieval term $\mathrm{e}^{\beta_{\text{net}} N \phi_1}$ and an $\boldsymbol{x}$-independent bulk $S \sim \mathrm{e}^{N[\alpha + g_{\max}(\beta_{\text{net}})]}$. At large N,

$$\beta_{\text{net}}^{-1} \ln[\mathrm{e}^{\beta_{\text{net}} N \phi_1} + S] \;\to\; \max(\phi_1,\ \phi_c(\alpha, \beta_{\text{net}})), \qquad \phi_c(\alpha, \beta_{\text{net}}) \;=\; \frac{\alpha + g_{\max}(\beta_{\text{net}})}{\beta_{\text{net}}}\,, \tag{10}$$

and the leading-order free energy per spin is

$$\frac{F(\phi_1; T, \alpha, \beta_{\text{net}})}{N} \;=\; -\max(\phi_1, \phi_c) \;-\; \tfrac{T}{2} \ln(1 - \phi_1^2). \tag{11}$$

The two branches meet in a cusp at $\phi_1 = \phi_c$. We call the right branch (slope -1) the *retrieval basin*: for $\phi_1 > \phi_c$ the retrieval term dominates the log-sum and the chain feels the linear pull of $\boldsymbol{\xi}^1$. We call the left branch (zero energy slope) the *saddle valley*: for $\phi_1 < \phi_c$ the bulk saddle dominates the log-sum, the energy in ϕ_1 is flat, and the sphere entropy $-(T/2)\ln(1 - \phi_1^2)$ alone pulls the chain toward $\phi_1 = 0$. Figure 2 plots (11) at $\alpha = 0.5$, $\beta_{\text{net}} = 1$, $\phi_c \approx 0.877$.

Single-pattern spinodal. The retrieval branch has a local minimum at $\phi_{\text{eq}}(T)$ from (3), valid only while $\phi_{\text{eq}}(T) > \phi_c$. As T rises, $\phi_{\text{eq}}(T)$ drops; when it reaches ϕ_c the local minimum disappears and the chain slides to $\phi_1 = 0$. The condition $\phi_{\text{eq}}(T) = \phi_c$ defines the single-pattern spinodal

$$\alpha_c^{\text{sp}}(\beta_{\text{net}}, T) \;=\; \beta_{\text{net}}\, \phi_{\text{eq}}(T) - g_{\max}(\beta_{\text{net}}), \qquad T_{\text{sp}}(\alpha, \beta_{\text{net}}) \;=\; \frac{\beta_{\text{net}}^2 - (\alpha + g_{\max}(\beta_{\text{net}}))^2}{\beta_{\text{net}}\,(\alpha + g_{\max}(\beta_{\text{net}}))}\,. \tag{12}$$

At $T = 0$, $\phi_{\text{eq}}(0) = 1$, and the two curves (9) and (12) share the same endpoint $(\beta_{\text{net}} - g_{\max}(\beta_{\text{net}}),\, 0)$.

Two boundaries, two protocols. At $T > 0$ the spinodal α_c^{sp} and the thermodynamic crossover α_c^{sd} separate, with $\alpha_c^{\text{sd}} < \alpha_c^{\text{sp}}$ (since $\alpha_c^{\text{sp}} - \alpha_c^{\text{sd}} = -(T/2)\ln(1 - \phi_{\text{eq}}^2) > 0$). In the strip $\alpha_c^{\text{sd}}(\beta_{\text{net}}, T) < \alpha < \alpha_c^{\text{sp}}(\beta_{\text{net}}, T)$ the basin still exists and lies above the cusp, but the bulk noise floor is lower in free energy. The escape barrier the chain must cross to leave the basin, $\Delta F_{\text{esc}}/N = (\phi_{\text{eq}} - \phi_c) + (T/2)\ln[(1 - \phi_{\text{eq}}^2)/(1 - \phi_c^2)]$, depends only on $\phi_{\text{eq}} - \phi_c$ and vanishes at α_c^{sp}, not at α_c^{sd}. The Kramers escape time $\tau \sim \exp(N\,\Delta F_{\text{esc}}/T)$ therefore grows exponentially with N throughout $\alpha < \alpha_c^{\text{sp}}$, regardless of which side of α_c^{sd} we are on. Consequently:

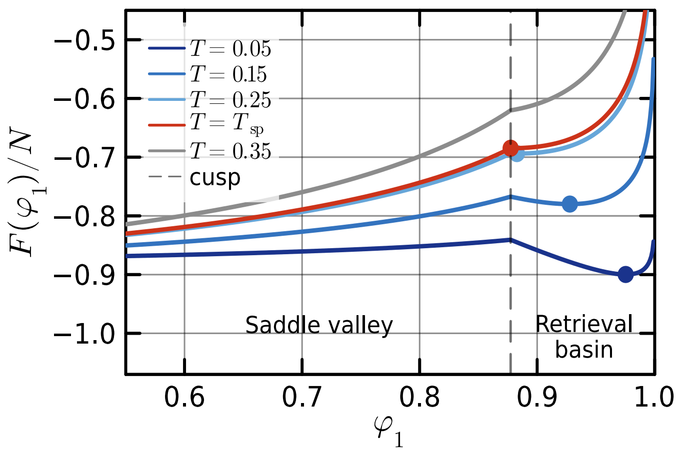

Figure 2: Leading-order free energy per spin $F(\phi_1)/N$ from (11) at $\alpha = 0.5$, $\beta_{\text{net}} = 1$, for five temperatures. The dashed vertical line marks the cusp $\phi_c = \alpha + g_{\max} \approx 0.877$ separating the saddle valley on the left from the retrieval basin on the right. Dots: basin minima $\phi_{\text{eq}}(T)$. At the spinodal temperature $T_{\text{sp}}(0.5, 1) \approx 0.262$ (defined in (12)) the basin minimum reaches the cusp; for $T = 0.35 > T_{\text{sp}}$ the basin no longer exists (grey curve).

- In the equilibrium (infinite-time) protocol, the chain Boltzmann-distributes between the basin and the noise floor; the empirical boundary is α_c^{sd}, the free-energy crossover.
- In the cold-initialised, finite-budget protocol (the practical retrieval scheme) the chain stays in the basin until its first Kramers escape. The asymptotic boundary at $N \to \infty$ is the spinodal α_c^{sp}, since $\tau \to \infty$ throughout $\alpha < \alpha_c^{\text{sp}}$. At finite N the empirical boundary is the Kramers contour $\tau(\alpha, T, N) = T_{\text{MC}}$, where T_{MC} is the simulation budget in MC steps, which lies inside the strip $\alpha_c^{\text{sd}} < \alpha < \alpha_c^{\text{sp}}$ and drifts toward α_c^{sp} as N grows.

At $T = 0$ both curves coincide at $\beta_{\text{net}} - g_{\max}(\beta_{\text{net}})$ and the distinction disappears. Figure 3 confirms the picture at $N = 100$: $\log_{10}\langle\tau\rangle$ at five α approaching the $T = 0$ endpoint 0.6226 shows the basin surviving the full budget 10^6 MC steps at low T and dropping by several orders of magnitude over a T window of width ~ 0.01 that slides toward $T = 0$ as α rises. A Kramers fit $\langle\tau\rangle \sim \tau_0\, \mathrm{e}^{c\Delta F/T}$, with τ_0 a prefactor and $c \in (0, 1]$ a barrier-reduction factor, predicts that the product $T \log\langle\tau\rangle$ reads off the effective barrier $c\Delta F$; in our data this product traces a shallow U in T at each α, with a finite high-T prefactor on one side and a T-dependent barrier on the other ($\phi_{\text{eq}}(T) - \phi_c$ shrinks as T rises). The minimum value of $T \log_{10}\langle\tau\rangle$ drops monotonically from ≈ 0.56 at $\alpha = 0.50$ to ≈ 0.16 at $\alpha = 0.58$ and below the measurement floor at $\alpha = 0.60$, consistent with the effective barrier vanishing as the spinodal is approached.

3 Monte Carlo schemes

We use three Metropolis variants on $\boldsymbol{x} \in \sqrt{N} \cdot S^{N-1}$, all at $\beta_{\text{net}} = 1$. They differ in how they treat the $M = \mathrm{e}^{\alpha N}$ patterns inside the LSE log-sum.

Direct. All M patterns are stored explicitly and the log-sum is evaluated over all of them at every step. Cost per step $\sim MN$. Direct MC is feasible up to $N \approx 25$ at $\alpha = 0.7$ ($M \approx 4 \times 10^7$); it is impossible at $N = 100$, $\alpha = 0.6$, where $M \approx 10^{26}$ exceeds available memory by twenty orders of magnitude.

Truncated. For each disorder draw we sample the $K \sim 10^4$ patterns whose alignment with $\boldsymbol{\xi}^1$ exceeds a cutoff ϕ_{keep}. The cutoff is chosen so that the expected count above it equals the

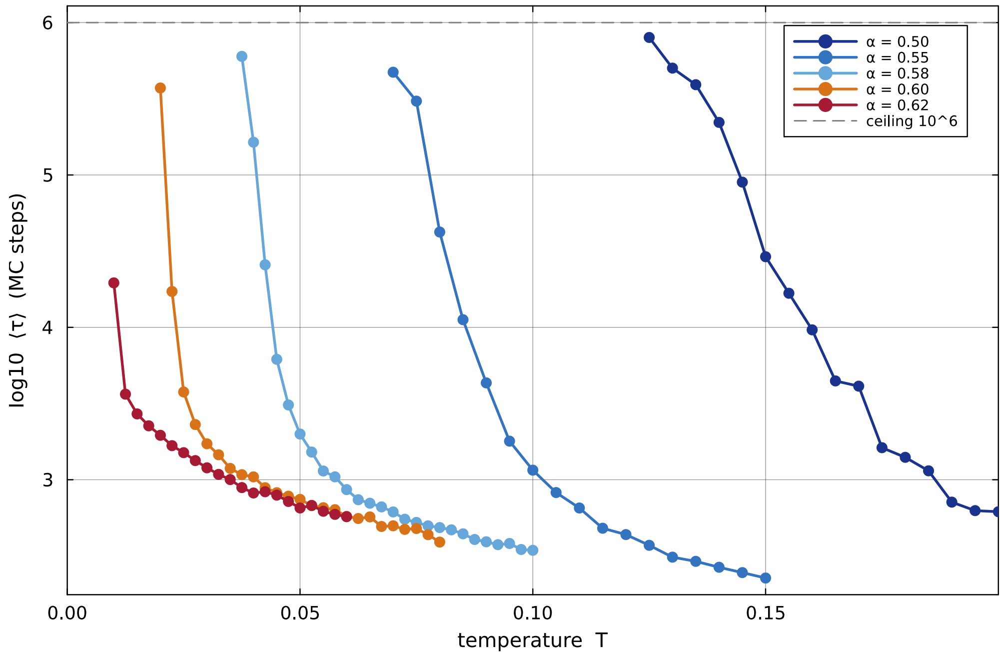

Figure 3: Mean basin escape time $\langle\tau\rangle$ versus T at $N = 100$ from N-dim saddle-bulk MC, 32 disorder samples per cell, $\beta_{\rm net} = 1$. Five values of α approaching $\alpha_c^{\rm sd}(1,0) \approx 0.6226$. Cells in which no disorder escaped within 10^6 MC steps are not shown; the grey dashed line marks that budget ceiling. The escape window narrows and slides toward $T = 0$ as α rises.

budget K. The remaining $M - K$ patterns below the cutoff enter the LSE log-sum through an analytic constant $\ln C_{\rm bulk} = \alpha N + \ln \int_{-1}^{\phi_{\rm keep}} h(\phi)\, \mathrm{e}^{N\phi}\, \mathrm{d}\phi$. The retained set is fixed once per disorder. At $N = 25$ truncated and direct produce the same basin boundary across the full α range we measure, so the discarded patterns are genuinely outside the basin physics rather than silently dropped. Truncated is the workhorse for the wide heatmap in Section 4.

Saddle-bulk. Same decomposition as truncated, with two changes: the cutoff is set at the cusp position $\phi_c(\alpha, \beta_{\rm net})$ from (10) rather than chosen by budget, and the bulk integral is replaced by its saddle-point value $M\, \mathrm{e}^{N g_{\max}(\beta_{\rm net})}$, an $\boldsymbol{x}$-independent constant inside the log-sum. Patterns above ϕ_c are drawn from the spherical density (2) restricted to $\phi > \phi_c$. In the rescaled variable $y = (1+\phi)/2 \in [0, 1]$, the spherical density becomes a symmetric Beta:

$$y \sim \mathrm{Beta}\!\left(\tfrac{N-1}{2}, \tfrac{N-1}{2}\right) \text{ truncated to } y \in \big[(1+\phi_c)/2,\, 1\big], \qquad \varphi_{1\mu} = 2y - 1. \tag{13}$$

Each retained pattern is then assembled from its $\varphi_{1\mu}$ and a uniform direction u_μ on the $(N-2)$-sphere perpendicular to $\boldsymbol{\xi}^1$. At the loads and dimension studied here ($N = 100$, $\alpha \in [0.5, 0.62]$), the expected count of alignments above ϕ_c is below unity for every α, so the algorithm typically retains only $\boldsymbol{\xi}^1$ explicitly and treats every other pattern through the bulk constant. This is a faithful sampler of the leading-order free energy (11): below the cusp the bulk constant inside the log-sum collapses $\beta_{\rm net}^{-1} \ln[\cdot]$ to the constant ϕ_c (the noise-floor branch), but the chain still feels the sphere geometry through the entropy term $-(T/2)\ln(1-\phi_1^2)$, which is what drives the slide to $\phi_1 = 0$.

***N*-reach and what each scheme tests.** Truncated at fixed $\phi_{\rm keep}$ keeps $K = M \Pr(\phi \geq \phi_{\rm keep}) \sim \mathrm{e}^{N[\alpha - I(\phi_{\rm keep})]}$ patterns, with $I(\phi) \equiv -\frac{1}{2}\ln(1-\phi^2)$ the large-N rate of the spherical tail. At $\phi_{\rm keep} = 0.4$, $I \approx 0.087$; for any $\alpha > 0.087$ the retained count grows exponentially in N and the GPU budget is exhausted by $N \sim 30$. Saddle-bulk sets $\phi_{\rm keep} = \phi_c(\alpha, \beta_{\rm net})$, which lies above the typical extreme $\phi_{\max}(\alpha) = \sqrt{1-\mathrm{e}^{-2\alpha}}$ of M inter-pattern alignments at every load below capacity, so the expected count $K = M\,\Pr(\phi \geq \phi_c) \to 0$ exponentially in N. With only $\boldsymbol{\xi}^1$ in the explicit sum, the spurious sum (4) collapses to $\max(\mathrm{e}^{\beta_{\mathrm{net}}N\phi_1}, M\,\mathrm{e}^{Ng_{\max}(\beta_{\mathrm{net}})})$ and the chain energy reduces to $H(\boldsymbol{x})/N \to -\max(\phi_1,\phi_c)$; together with the sphere entropy $-(T/2)\ln(1-\phi_1^2)$ this reproduces the leading-order free energy (11) on the N-sphere.

The two schemes therefore probe different physics. Saddle-bulk realises the Kramers kinetics of (11) at large N, but the analytic bulk $M\,\mathrm{e}^{Ng_{\max}(\beta_{\mathrm{net}})}$ is taken on faith and any correction to $g_{\max}$ would be invisible. Truncated with ϕ_{keep} near or below the integrand saddle $\phi^*(\beta_{\mathrm{net}})$ retains the patterns that dominate the spurious-sum integral and audits the bulk directly; a subleading correction to $g_{\max}$ would appear as a mismatch between truncated and direct boundaries. At $N = 25$ the two agree across our α range, so the leading-order analytic bulk captures the spurious physics to within MC noise at the boundary.

Why local refresh fails. A faster scheme (sometimes called “smart”) refreshes the retained set every N_{refresh} steps from a cone around the current chain position, keeping only the patterns whose alignment with $\boldsymbol{x}$ is currently large. The refresh breaks the quenched-disorder assumption that underlies (3)–(9). As $\boldsymbol{x}$ wanders, new patterns are minted at every refresh, so the cumulative set of patterns seen along a single trajectory exceeds M, and the chain effectively faces a denser pattern set than the model defines. In the retrieval basin the spurious-recruitment rate alone destroys retrieval, for reasons unrelated to (9). Truncated and saddle-bulk both fix the retained set once per disorder and avoid the issue.

4 Results

Figure 4 shows $\langle\phi_1\rangle - \phi_{\mathrm{eq}}(T)$ over the (α, T) plane at $\beta_{\mathrm{net}} = 1$. The bulk of the plane, $\alpha \in [0.20, 0.70]$, uses truncated MC at varying N (the load $M = \mathrm{e}^{\alpha N}$ is held fixed at $M \approx 4.4\times10^6$ by ramping N with α). A fine T-grid saddle-bulk MC at $N = 100$ fills in the rectangle $\alpha \in [0.5, 0.62]$, $T \in [0, 0.10]$.

The basin region (white) is bounded by a sharp dark-red transition that pinches at $\alpha \approx 0.6226$ at $T = 0$. Ramsauer’s $\alpha_c^{\mathrm{G}} = 1/2$ from (5) and the exact-edge curve α_c^{bd} from (6) both sit visibly off the empirical boundary at low T. The thermodynamic crossover α_c^{sd} from (9) (red) and the spinodal α_c^{sp} from (12) (purple) bound a strip $\alpha_c^{\mathrm{sd}} < \alpha < \alpha_c^{\mathrm{sp}}$ at $T > 0$, sharing the single endpoint at $T = 0$. The empirical line at $N = 100$ sits inside this strip, on the Kramers contour $\tau(\alpha, T, N) = T_{\mathrm{MC}}$ where the chain escapes within the simulation budget. As N grows the contour drifts toward α_c^{sp}, the cold-initialised asymptotic boundary discussed in §2.

Finite-N residual basin above the LO boundary. At $T \to 0$ the empirical boundary in Figure 4 extends past $\alpha_c^{\mathrm{sd}} = \alpha_c^{\mathrm{sp}}(T=0) = 0.6226$ to $\alpha = 0.70$. The $N(\alpha) = \lceil \log M/\alpha \rfloor$ ramp at $M = 4.4\times10^6$ gives $N = 25, 24, 22$ at $\alpha = 0.62, 0.65, 0.70$ respectively, where the LO limit is not yet reached. The LO corner $\max(\phi_1, \phi_c)$ in (11) is the $N \to \infty$ limit of

$$\frac{V_N(\phi_1)}{N} = -\frac{1}{\beta_{\mathrm{net}}N}\log\left(\mathrm{e}^{\beta_{\mathrm{net}}N\phi_1} + \mathrm{e}^{N(\alpha+g_{\max})}\right) - \tfrac{T}{2}\log(1-\phi_1^2), \tag{14}$$

which rounds the corner over a ϕ_1-width $1/(\beta_{\mathrm{net}}N)$. At the three loads the rounding width is $0.040, 0.042, 0.045$, and the gap $\phi_c - 1$ is $-0.003, +0.028, +0.078$. The rounding is therefore comparable to or larger than the gap. Figure 5 plots (14) against the LO curve at $T = 0.005$. Each panel shows a local minimum of V_N at ϕ_1 close to 1 that the LO curve does not have.

Numerically, the local minimum of V_N sits at $\phi_1 = 0.995, 0.992, 0.973$ at $\alpha = 0.62, 0.65, 0.70$, with $[V_N(\phi_1^{\mathrm{well}}) - V_N(0)]/N = -1.50\times10^{-2}, -4.50\times10^{-3}, +2.9\times10^{-3}$ respectively. The cold-initialised $\langle\phi_1\rangle$ at $T = 0.005$ in the same α columns of Figure 4 reads $0.9946, 0.9910, 0.9753$, within 0.002 of the predicted local-minimum position in each case.

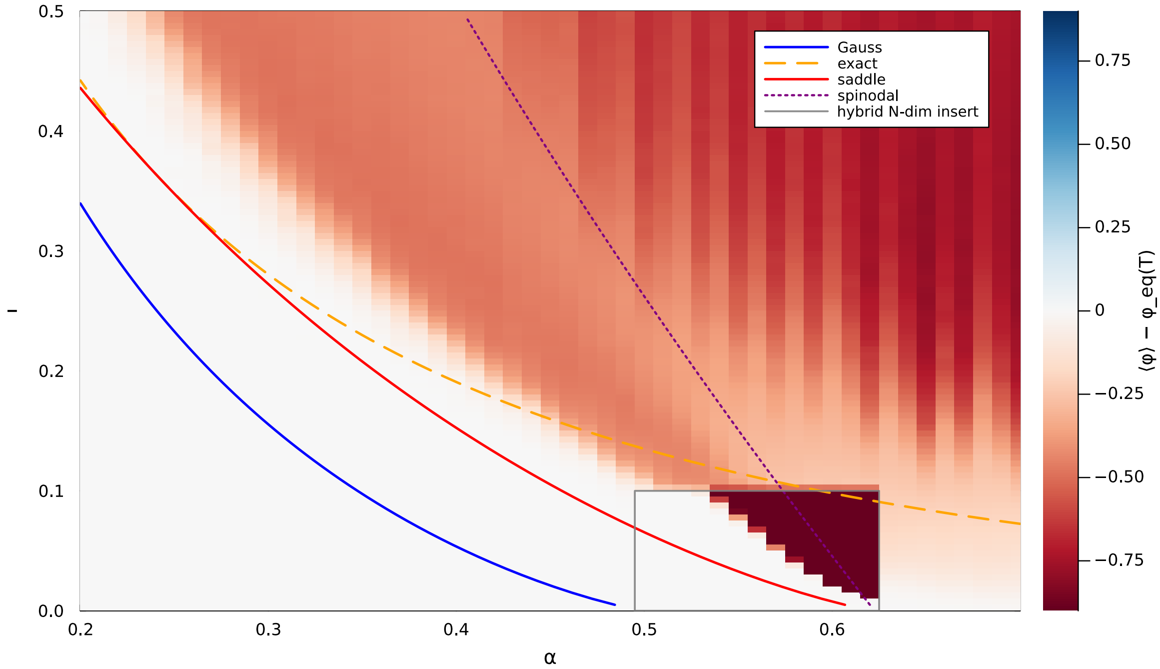

Figure 4: $\langle\phi_1\rangle - \phi_{\rm eq}(T)$ from Monte Carlo over the (α, T) plane at $\beta_{\rm net} = 1$. Background: truncated MC at varying N ($\alpha \in [0.20, 0.70]$, $M \approx 4.4 \times 10^6$). Grey rectangle: N-dim saddle-bulk MC at $N = 100$ on a fine T grid. Curves: $\alpha_c^{\rm G}$ (5) (blue solid, Gaussian + support edge); $\alpha_c^{\rm bd}$ (6) (orange dashed, exact density + support edge); $\alpha_c^{\rm sd}$ (9) (red solid, thermodynamic F-crossover); $\alpha_c^{\rm sp}$ (12) (purple dotted, single-pattern spinodal). $\alpha_c^{\rm sd}$ and $\alpha_c^{\rm sp}$ share the endpoint at $T = 0$ where the basin disappears; the empirical line at finite N sits between them, on the Kramers contour, drifting toward $\alpha_c^{\rm sp}$ as N grows.

For $\alpha = 0.62$ and $\alpha = 0.65$ the local minimum is the global minimum of V_N; the residual basin survives thermodynamically and the chain settles in it independently of dynamics. For $\alpha = 0.70$ the local minimum lies above the bulk floor; the residual basin is metastable. The barrier of V_N traverses a saddle at $\phi_1 \approx 0.92$ with $V_N/N = -1.07415$, and the per-chain exponent $N[V_N(\phi_1^{\rm saddle}) - V_N(\phi_1^{\rm well})]/T = 22 \cdot 3.0 \times 10^{-4}/0.005 \approx 1.3$ is small in Boltzmann units. Empirically, no chain in the 32-disorder $\times$ 2-replica ensemble escapes the residual well within the $2^{15} + 2^{13}$ MC steps of the run. We do not claim a Kramers escape time from V_N alone, because the disorder-averaged V_N does not include the realisation-specific landscape roughness that may also contribute to trapping.

The residual basin is a finite-N artefact. Solving $dV_N/d\phi_1 = 0$ for the local minimum of V_N in (14), the analytic minimum exists at $\alpha = 0.70$, $\beta_{\rm net} = 1$, $T = 0.005$ for $N \le 23$ and disappears for $N \ge 24$. The $M = 3 \times 10^8$ Nramp from §3 gives $N(0.65) = 30$ and $N(0.70) = 28$, so V_N loses its local minimum at $\alpha = 0.70$ and the empirical boundary there should agree with the LO landscape. At $\alpha = 0.65$ the local minimum persists at $N = 30$, at $\phi_1 = 0.990$ with $[V_N(\phi_1^{\rm well}) - V_N(0)]/N = +4.0 \times 10^{-4}$ (metastable, per-chain exponent 2.4 at $T = 0.005$); the residual basin shrinks but does not yet disappear. The empirical boundary in that data is the planned diagnostic.

5 Discussion

The Ramsauer threshold $\alpha_c = 1/2$ is a Gaussian artefact. The Gaussian density misses the interior saddle that the exact spherical density places at $\phi^*(\beta_{\rm net})$ in the partition-sum integrand; restoring it gives a finite $\beta_{\rm net}$-dependent capacity $\alpha_c^{\rm sd}(\beta_{\rm net}, 0) = \alpha_c^{\rm sp}(\beta_{\rm net}, 0) = \beta_{\rm net} - g_{\max}(\beta_{\rm net})$, the load at which the cusp $\phi_c = (\alpha + g_{\max})/\beta_{\rm net}$ reaches the unit sphere and the retrieval basin disappears.

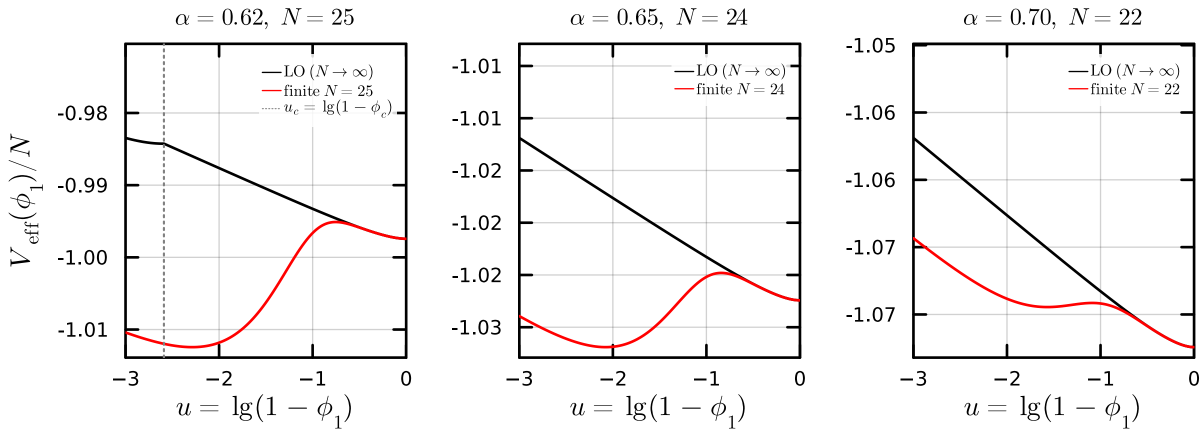

Figure 5: Effective free energy $V_{\text{eff}}(\phi_1)/N$ at $\beta_{\text{net}} = 1$, $T = 0.005$, against $u = \lg(1 - \phi_1)$ for three loads above $\alpha_c^{\text{sd}} = 0.6226$. Black: leading order, $V_{\text{LO}}/N = -\max(\phi_1, \phi_c) - (T/2)\log(1-\phi_1^2)$ from (11). Red: the finite-N form V_N/N from (14), at the $N(\alpha)$ from the $M = 4.4 \times 10^6$ Nramp. The smoothing of the LO corner over a width $1/(\beta_{\text{net}}N)$ generates a local minimum at ϕ_1 close to 1 that the LO curve does not have. Grey dotted: LO cusp position $u_c = \lg(1-\phi_c)$ (only inside the window for $\alpha = 0.62$).

The capacity sits below Ramsauer's bound at moderate β_{net}, crosses it at $\beta_{\text{net}} \approx 0.7$, and grows logarithmically beyond. The β_{net}-independence of Ramsauer's $1/2$ is itself the signature of the right-edge replacement: at the edge the β_{net}^{-1} prefactor of H and the β_{net} inside the exponent cancel, while at the interior saddle the cancellation breaks through the β_{net}-dependence of ϕ^*.

At $T > 0$ the saddle line α_c^{sd} and the spinodal α_c^{sp} separate. α_c^{sd} is the thermodynamic boundary at which the basin and the saddle valley exchange order in free energy; α_c^{sp} is the geometric boundary at which the cusp catches the basin minimum and the retrieval branch disappears. For a cold-initialised chain the cusp is the only escape route, with barrier $(\phi_{\text{eq}} - \phi_c) + (T/2)\ln[(1-\phi_{\text{eq}}^2)/(1-\phi_c^2)]$ vanishing only at α_c^{sp}. The Kramers escape time grows exponentially in N throughout $\alpha < \alpha_c^{\text{sp}}$, irrespective of which side of α_c^{sd} the chain sits on; at $N \to \infty$ the empirical boundary is therefore the spinodal. At finite N it is the Kramers contour inside the strip $\alpha_c^{\text{sd}} < \alpha < \alpha_c^{\text{sp}}$, which is what our saddle-bulk MC at $N = 100$ measures.

Smoothed bulk and the LSR resemblance. By spherical isotropy and at large N the bulk smoothed field $\langle S(\boldsymbol{x})\rangle = M\,\mathrm{e}^{Ng_{\max}(\beta_{\text{net}})}$ is $\boldsymbol{x}$-independent on the unit sphere: there is no softening length to insert because there is no $\boldsymbol{x}$-dependence to regularize. Inside the log-sum-exp this collapses the spurious branch of $H(\boldsymbol{x})/N$ to the constant $-\phi_c$, and only the spherical Jacobian $-(T/2)\ln(1-\phi_1^2)$ drives the chain below the cusp. The same flat-bulk dynamics appears in the compactly supported LSR kernel of Petrova et al. [8]: below its support cutoff at $\phi_{\text{cut}} = 1 - 1/b$ the kernel returns no force on the chain. Two consequences are shared between the models: (i) the spurious-phase one-point distribution below the threshold is the bare spherical density $\propto (1-\phi_1^2)^{(N-3)/2}$, T-independent, because the potential is flat there and only the sphere Jacobian sets $P(\phi_1)$; (ii) the spurious-side contribution to the Kramers barrier has the same entropy-only form $(T/2)[\ln(1-\phi_{\text{threshold}}^2) - \ln(1-\phi_{\text{eq}}^2)]$. The two models differ on the α-dependence of the threshold: $\phi_c(\alpha, \beta_{\text{net}})$ slides upward with α in LSE, while $\phi_{\text{cut}}(b)$ is α-independent in LSR. Capacity in LSE is set by the cusp reaching the unit sphere; in LSR capacity comes from a kernel-specific mechanism. The stability maps therefore share bulk-side dynamics but not level curves in the (α, T) plane.

What the saddle-bulk run validates. The truncated scheme at $N = 25$ audits the LO bulk replacement directly: patterns retained above ϕ_{keep} overlap with the spurious saddle, and agreement with the direct simulator across the α range is a positive test of the analytic bulk. At

$N = 100$ no individual pattern survives above the cusp, so the saddle-bulk scheme retains only $\boldsymbol{\xi}^1$ and integrates the leading-order Kramers kinetics on (11) directly. Agreement with the analytic boundary at $N = 100$ confirms the LO kinetics in the regime where direct sampling is infeasible, but a subleading correction to $g_{\max}(\beta_{\text{net}})$ would shift all four boundary curves uniformly and remain invisible to a saddle-bulk-only check. The LO bulk replacement therefore rests on the $N = 25$ truncated–vs–direct comparison; the saddle-bulk extends the verification to the Kramers regime at large N.

References

[1] Daniel J. Amit, Hanoch Gutfreund, and Haim Sompolinsky. Statistical mechanics of neural networks near saturation. *Annals of Physics*, 173(1):30–67, 1987.

[2] Mete Demircigil, J. Heusel, Matthias Löwe, S. Upgang, and F. Vermet. On a model of associative memory with huge storage capacity. *Journal of Statistical Physics*, 168(2):288–299, 2017.

[3] John J. Hopfield. Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences*, 79(8):2554–2558, 1982. doi: 10.1073/pnas.79.8.2554.

[4] Dmitry Krotov and John J. Hopfield. Dense associative memory for pattern recognition. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 29, pages 1172–1180, 2016.

[5] Dmitry Krotov and John J. Hopfield. Large associative memory problem in neurobiology and machine learning. In *International Conference on Learning Representations*, 2021.

[6] Tatiana Petrova. Thermal robustness of retrieval in dense associative memories: LSE vs LSR kernels. In *New Frontiers in Associative Memory Workshop at ICLR 2026*, 2026. arXiv:2603.13350.

[7] Tatiana Petrova, Evgeny Polyachenko, and Radu State. Geometric entropy and retrieval phase transitions in continuous thermal dense associative memory. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, Proceedings of Machine Learning Research. PMLR, 2026. arXiv:2604.07401.

[8] Tatiana Petrova, Evgeny Polyachenko, and Radu State. Retrieval metastability via Kramers escape and centroid hopping. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. placeholder key; replace at submission.

[9] Hubert Ramsauer, Bernhard Schäfl, Johannes Lehner, Philipp Seidl, Michael Widrich, Lukas Gruber, Markus Holzleitner, Thomas Adler, David Kreil, Michael K. Kopp, Günter Klambauer, Johannes Brandstetter, and Sepp Hochreiter. Hopfield networks is all you need. In *International Conference on Learning Representations*, 2021.